

Leadership Initiatives

AI & Emerging Technology Internship



Machine Learning & Artificial Intelligence for Business: Key Concepts and Trends

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July 27, 2026



Slides: <https://tinyurl.com/EspinosaLead>



PAML4M Book



[Amazon](#)
[GitHub](#)



[Ch1](#)



R:



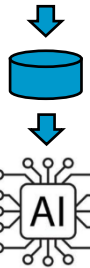
KOGOD SCHOOL of BUSINESS
AMERICAN UNIVERSITY • WASHINGTON, DC

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Prof. Espinosa holds a [Ph.D.](#) and [Master of Science in Information Systems](#) from the [Tepper School of Business](#) at [Carnegie Mellon University](#), a [Master of Business Administration](#) from [Texas Tech University](#), and a [Mechanical Engineering](#) degree from [Pontificia Universidad Catolica del Peru](#). He is the architect and implementor of [Kogod's MS Analytics Program](#) and the undergraduate specialization in [Business Analytics](#). He is also the curriculum architect for the [IT&A programs](#) and specializations. He has over 70 published journal articles, conference proceedings, books, and book chapters. He has co-authored two research books, one on work coordination across time zones and a 2-volume book on big data and analytics for service delivery. He has also written a book titled [“Predictive Analytics and Machine Learning for Managers”](#) with practical applications in R. His research focuses on coordination and performance in technical projects across global boundaries, mainly distance and time zones. More recently, he has developed social network and graph analytics methods to represent team knowledge quantitatively and visually. He has 70 peer-reviewed articles, many of them published in leading scholarly journals, including [Management Science](#), [Organization Science](#), [Information Systems Research](#), the [Journal of Management Information Systems](#), [IEEE Transactions on Software Engineering](#), [IEEE Transactions on Engineering Management](#), [Communications of the ACM](#), [Human Factors, Information, Technology and People](#), and [Software Process: Improvement and Practice](#). His work has also been featured in leading academic conference proceedings. He teaches predictive analytics with machine learning, social & organizational network analytics, R programming, information technology foundations, business process analysis, and programming for business applications. He has received multiple service, research, and teaching awards, including [“American University 2025 Outstanding Teacher”](#), [“2022 Kogod Graduate Professor of the Year”](#), and [“American University 2022 Green Teacher”](#). He has several years of work experience, first as a mechanical design engineer and later as a senior manager, VP, and CFO with international organizations, directly supporting, supervising, and formulating policy for finance and global IT, data management, and analysis, where he designed and developed many software applications to support geographically distributed work.

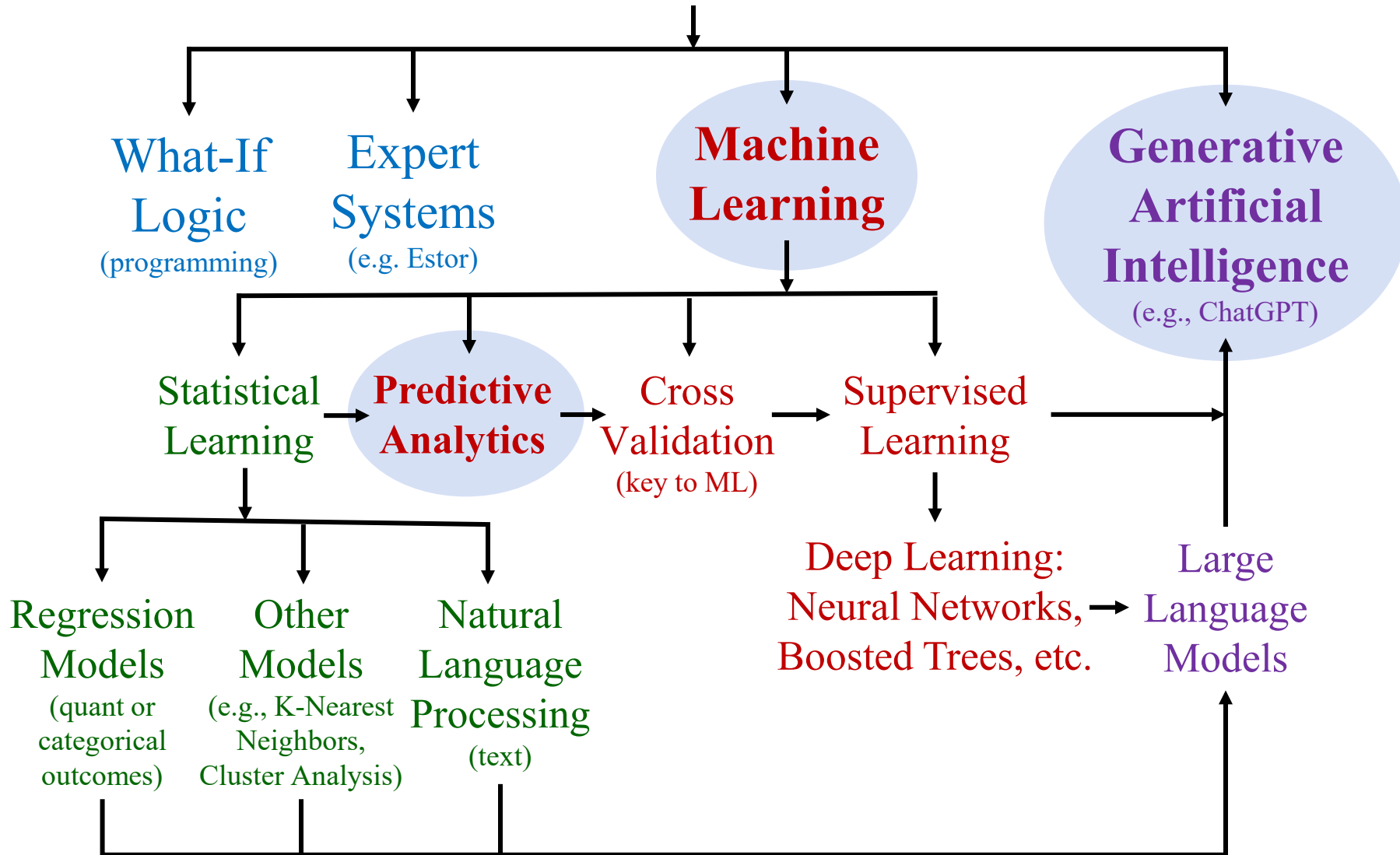
A Story about Estor – AI is Not New

- 🗨️ **What was Estor?** It is one of the earliest expert systems applied to software effort estimation, developed by Steven Vicinanza, a Carnegie Mellon PhD student in 1990, and published in MIS Quarterly in June of 1992, “*Examining the Feasibility of a Case-Based Reasoning Model for Software Effort Estimation*”
- It was built by analyzing how **experienced** software project estimators reasoned when estimating time to completion (i.e., effort) in real software projects.
- ⚙️ **How did it work?** When estimating new projects, these experts retrieved similar past cases and applied adaptation **if-then rules**. These rules were coded into a **rule database** to be queried by the **Expert System**, mirroring the expert’s thought process.
- 📊 **Results:** Estor was tested on 15 real software projects, comparing its estimates against those of the experts and various quantitative estimation methods. **Estor** was as accurate as the human experts and more accurate than the traditional estimation methods.
- ⚙️ **Importance:** Demonstrated the feasibility of expert system AI for software estimation back in the **early 1990s**, when machine learning was not mainstream, which inspired over three decades of follow-up research using machine learning, neural networks, regression, and fuzzy systems to improve estimation accuracy.



In summary, **Estor** was a groundbreaking early **AI software effort estimator** using insights from expert estimators. It proved that capturing **expert thinking** in a structured rule base could match human-level estimates, setting the stage for decades of research in this field.

Main Branches of AI





Statistical Learning

Predictive Analytics

Cross Validation (key to ML)

Supervised Learning

Predictive Analytics

THE WALL STREET JOURNAL
WSJ.com

September 11, 2015, 9:28 AM ET

Facebook 'Likes' Mean a Computer Knows You Better Than Your Mother

By Georgia Wells



Multiple studies over the years have confirmed that “likes” can predict traits, preferences, personality, etc., with 85% to 95% accuracy.

A computer needs the data from only 10 Facebook “likes” to beat the accuracy of a person’s coworker at judging his or her personality traits, such as extraversion, conscientiousness and neuroticism. The computer needs 70 “likes” to be more accurate than a person’s friends. With 250 Facebook data points, the computer can beat someone’s spouse.



Business Analytics

“It is the scientific process of transforming data into insights for making better decisions” (INFORMS)

Lack analytical talent – by **2018**, the US would face a shortage of **140,000** to **190,000** people with **deep analytical skills** and **1.5 million managers** with analytical skills to make decisions with data (McKinsey Global Institute, **2011**)



The Age of Analytics
McKinsey Global Institute
2016



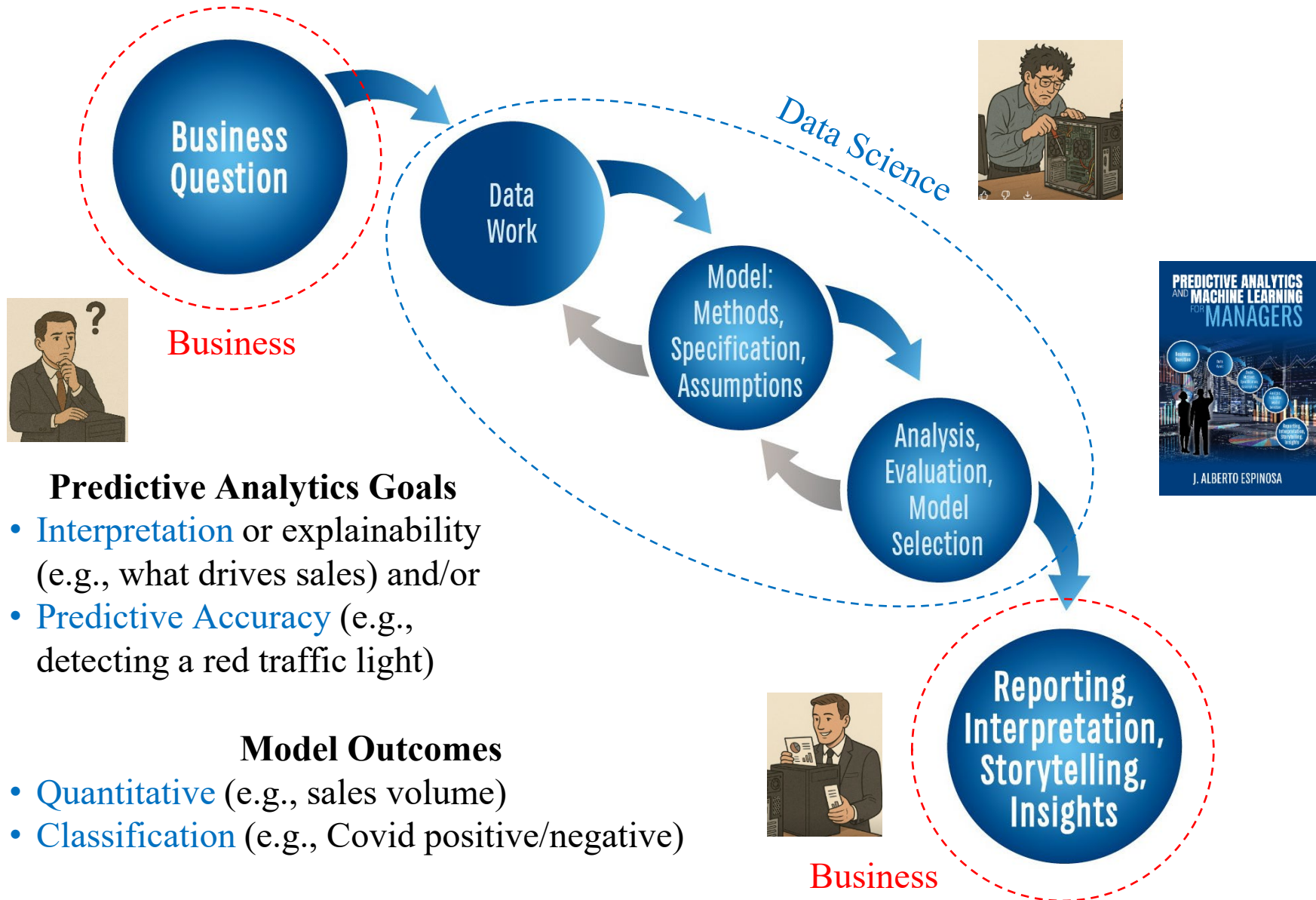
2M-4M

Projected US demand for
“**business translators**” in
the **2020** decade

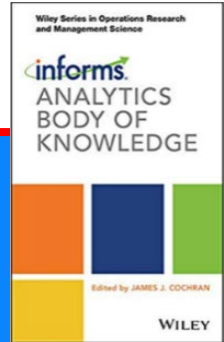
High demand for **business translators** → people who serve as the **link** between **data science** talent and business applications and **managers**, who are business-savvy and can answer business questions.

In **2025** McKinsey issued their “**AI in the Workplace**” report, stating that employers are eager to adopt AI, but leadership and readiness in this area lag behind → Almost **all** companies invest in **AI**, but just **1%** believe they lack **leaders** who understand AI.

The Predictive Analytics Process



Analytics Body of Knowledge



Business Domains

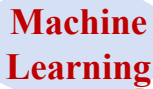
Business
Healthcare
Marketing
Social Networks
Accounting, Fraud Detection & Forensics
Financial Analysis
Data Governance
Sports
Cyber Security

Analytics & Data Science

Descriptive Analytics
Prescriptive Analytics
Text Analytics
Large Language Models
Artificial Intelligence
Predictive Analytics
Machine Learning
Business Intelligence
Data Visualization

Foundations (IT & Math)

Information Technology
Database & Big Data
Software & Tools
Data Warehousing
Math & Statistics
Programming R, Python

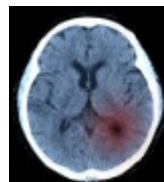
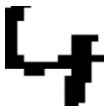


Machine Learning

Is method of data analysis that uses algorithms that **learn from data** iteratively, allowing computers to find hidden insights **without being explicitly programmed**

(SAS: http://www.sas.com/en_us/insights/analytics/machine-learning.html)

- **Unsupervised learning** → data exploration **without** specific **goals** (closely associated with descriptive analytics and data mining, e.g., clustering, correlation analysis, histograms)
- **Supervised learning** → data analysis **with** specific **goals** in mind (closely associated with analytics, e.g., regression) → not much different than **predictive modeling**

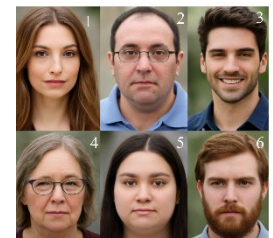


\$600



\$200 OFF
Miele CM7750 Fully
Automatic Espresso Machine

\$5,300



Classify each person as “hot” or “not” to you:



LooksMapping app by a 22-year old developer to rate restaurants on a “hotness” scale

Predictive Analytics Model

Without Machine Learning or Cross Validation

Predictors



Outcome

PizzaID	brand	mois	prot	fat	ash	sodium	carb
14001	D	47.17	22.29	21.3	4.08	0.74	5.16
14002	D	49.16	27.99	17.49	3.29	0.39	2.07
14003	A	30.49	21.28	41.65	4.82	1.64	1.76
14004	B	52.68	14.38	25.72	3.26	0.93	3.96
14005	H	33.05	7.34	15.78	1.34	0.42	42.49
14006	H	35.55	7.32	16.4	1.76	0.36	38.97
14007	G	28.68	8.3	16.07	1.41	0.45	45.54

cal
302
278
467
305
341
333
360

Machine Learning w/Cross-Validation

→ Split data set + Train model + Test it

Training Subset: *Train Model* → *Outcome*

PizzaID	brand	mois	prot	fat	ash	sodium	carb
14001	D	47.17	22.29	21.3	4.08	0.74	5.16
14002	D	49.16	27.99	17.49	3.29	0.39	2.07
14003	A	30.49	21.28	41.65	4.82	1.64	1.76
14004	B	52.68	14.38	25.72	3.26	0.93	3.96

cal
302
278
467
305

Test Subset: *Test Model's Accuracy* →

14005	H	33.05	7.34	15.78	1.34	0.42	42.49
14006	H	35.55	7.32	16.4	1.76	0.36	38.97
14007	G	28.68	8.3	16.07	1.41	0.45	45.54

Outcome

341	290	51
333	362	-29
360	371	-11

Actual Predicted Error
Mean Squared Error: 1187.7
RMSE = 34.5

Selected Model

New Data

D	49.16	27.99	17.49	3.29	0.39	2.07
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*Uses: test and compare models, tune model parameters,
train deep and supervise learning models, etc.*

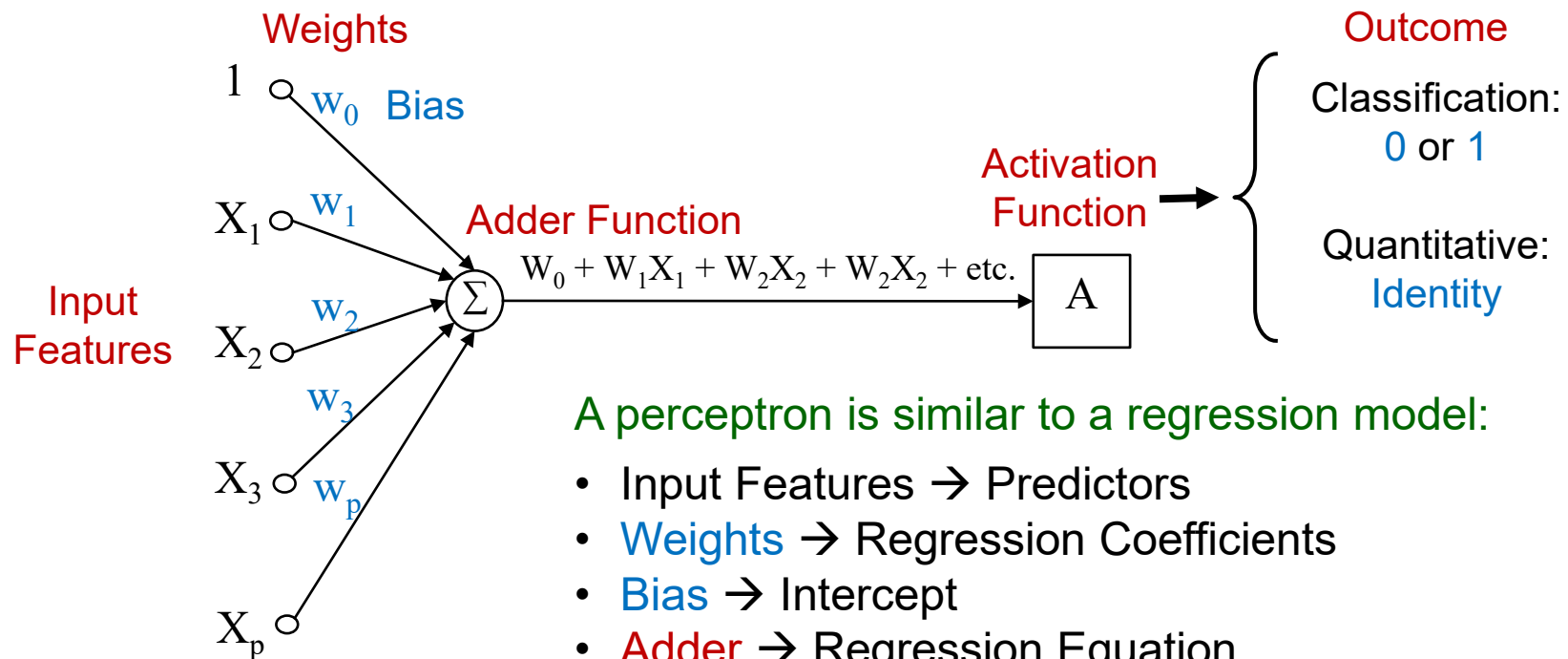


Predict:

- Values
- Classifications
- Words
- Images, etc.

The Simplest Neural Network → The Perceptron

The **perceptron** is the simplest model of a **computational neuron** and constitutes the **basic component** of a **neural network**. All neural networks are composed of one or many perceptrons. A perceptron contains **input features**, **weights**, an **adder**, an **activation function** and an **output**.



A perceptron is similar to a regression model:

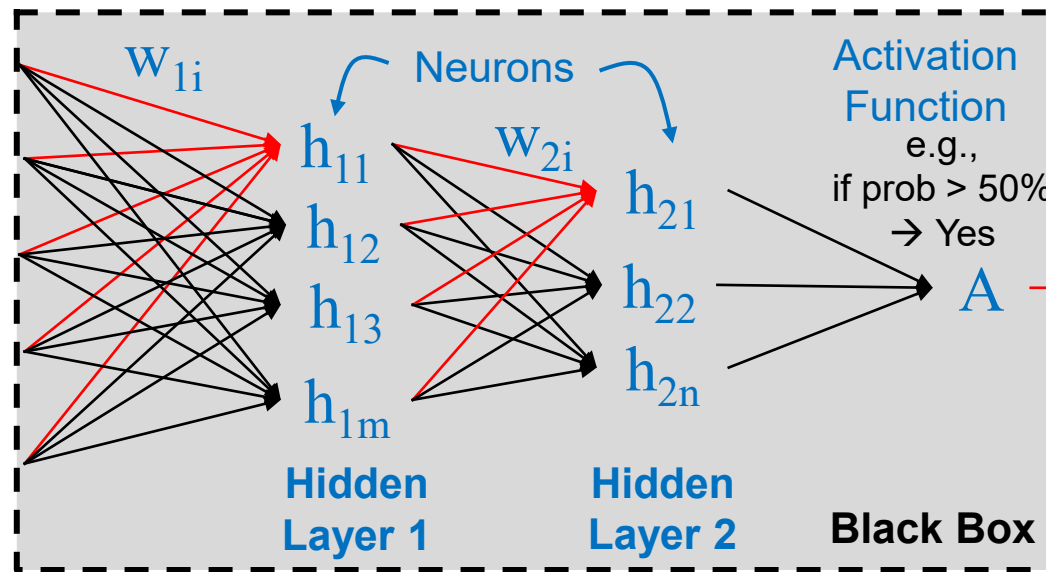
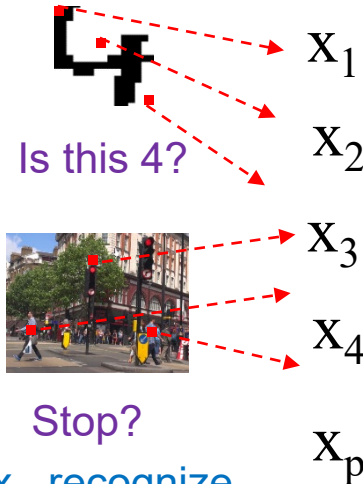
- Input Features → Predictors
- **Weights** → Regression Coefficients
- **Bias** → Intercept
- **Adder** → Regression Equation
- **Activation Function** →
 - ✓ **Sigmoid** (reverse Logit), if classification
 - ✓ **Identity**, if quantitative

Machine Learning Models (e.g., Neural Networks)

- **Neural Network** → a collection of connected perceptrons
- **Simple learning** → simple linear relationships & 1 or 2 layers
- **Deep learning** → linear or non-linear relationships & multiple layers

ex. recognize
hand-written
zip code
numbers

**Input
Layer**

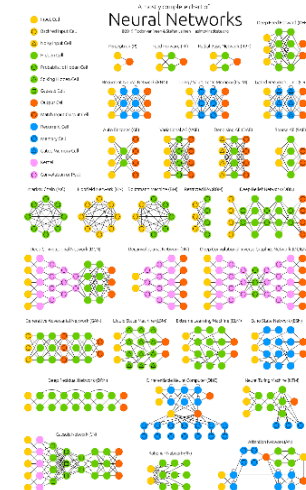
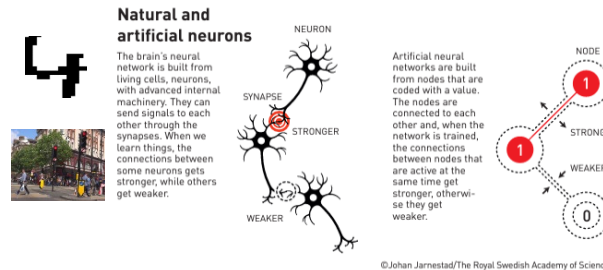


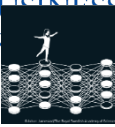
Yes

y

Outcome

- Values
- Classifications
- Words
- Images, etc.





Neural Networks

Biological



Artificial

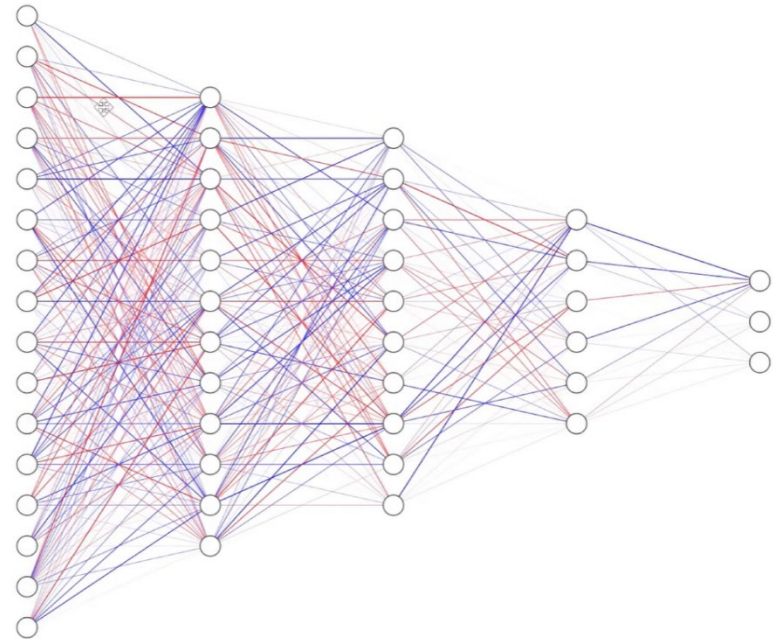
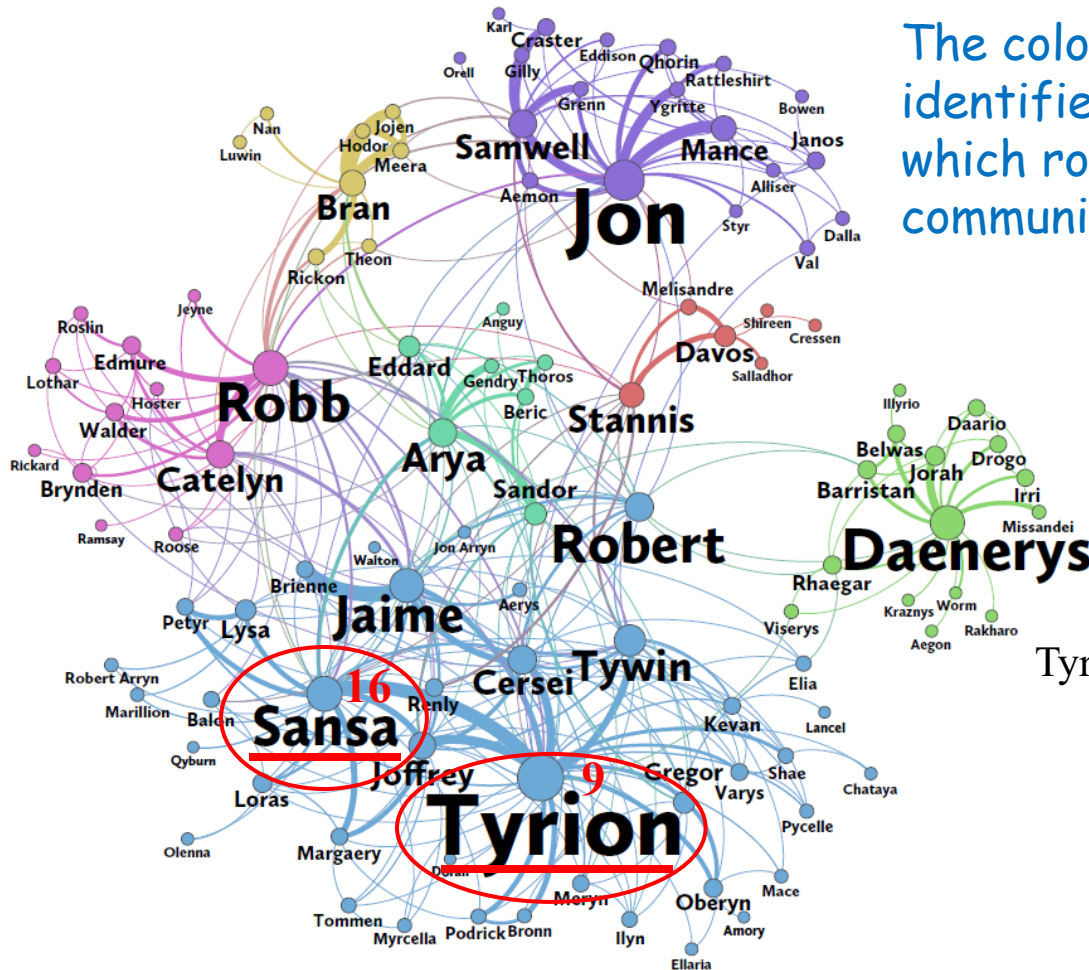


Illustration: Network of Thrones

Tie weighs: the number of times the pair appeared within 15 words in the GoT book

The colors depict communities identified with network analytics, which roughly coincide with the communities in the book.



Sansa

Tyrion

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1		0	0	3	8	8	8	3	8	8	8	3	0	0	3	3
2	0		0	3	8	8	8	3	8	8	8	3	0	0	3	3
3	0	0		3	8	8	8	3	8	8	8	3	0	0	3	3
4	3	3	3		3	3	3	0	3	3	3	0	3	3	8	8
5	8	8	8	3		0	0	3	0	0	0	3	8	8	11	11
6	8	8	8	3	0		0	3	0	0	0	3	8	8	11	11
7	8	8	8	3	0	0		3	0	0	0	3	8	8	11	11
8	3	3	3	0	3	3	3		3	3	3	0	3	3	8	8
9	8	8	8	3	0	0	0	3		0	0	3	8	8	11	11
10	8	8	8	3	0	0	0	3	0		0	3	8	8	11	11
11	8	8	8	3	0	0	0	3	0	0		3	8	8	11	11
12	3	3	3	0	3	3	3	0	3	3	3		3	3	8	8
13	0	0	0	3	8	8	8	3	8	8	8	3		0	3	3
14	0	0	0	3	8	8	8	3	8	8	8	3	0		3	3
15	3	3	3	8	11	11	11	8	11	11	11	8	3	3		0
16	3	3	3	8	11	11	11	8	11	11	11	8	3	3	0	

Source: <https://networkofthrones.com>

Illustration: Social network constructed with text and visual analytics of the Bible (IBM)

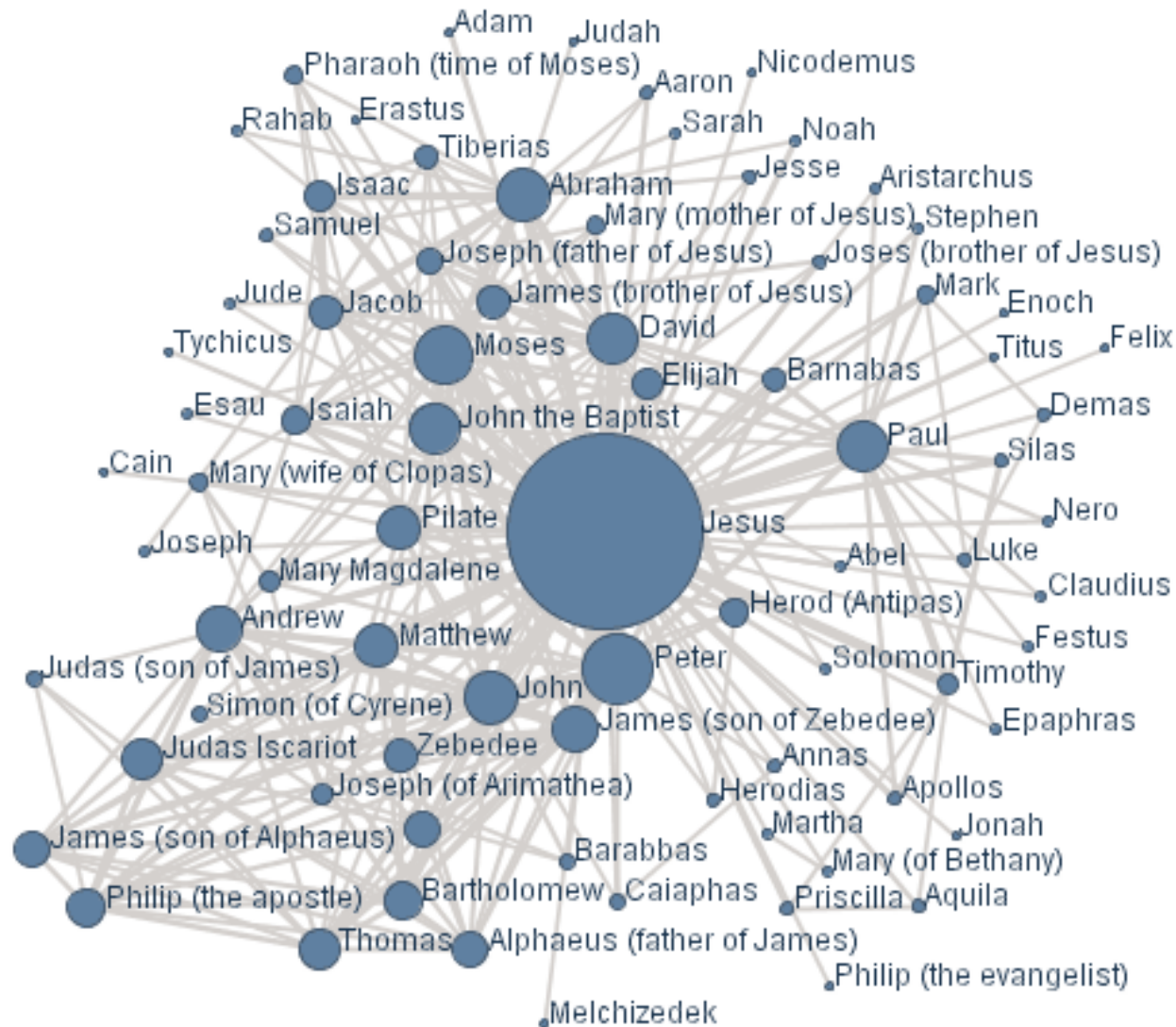
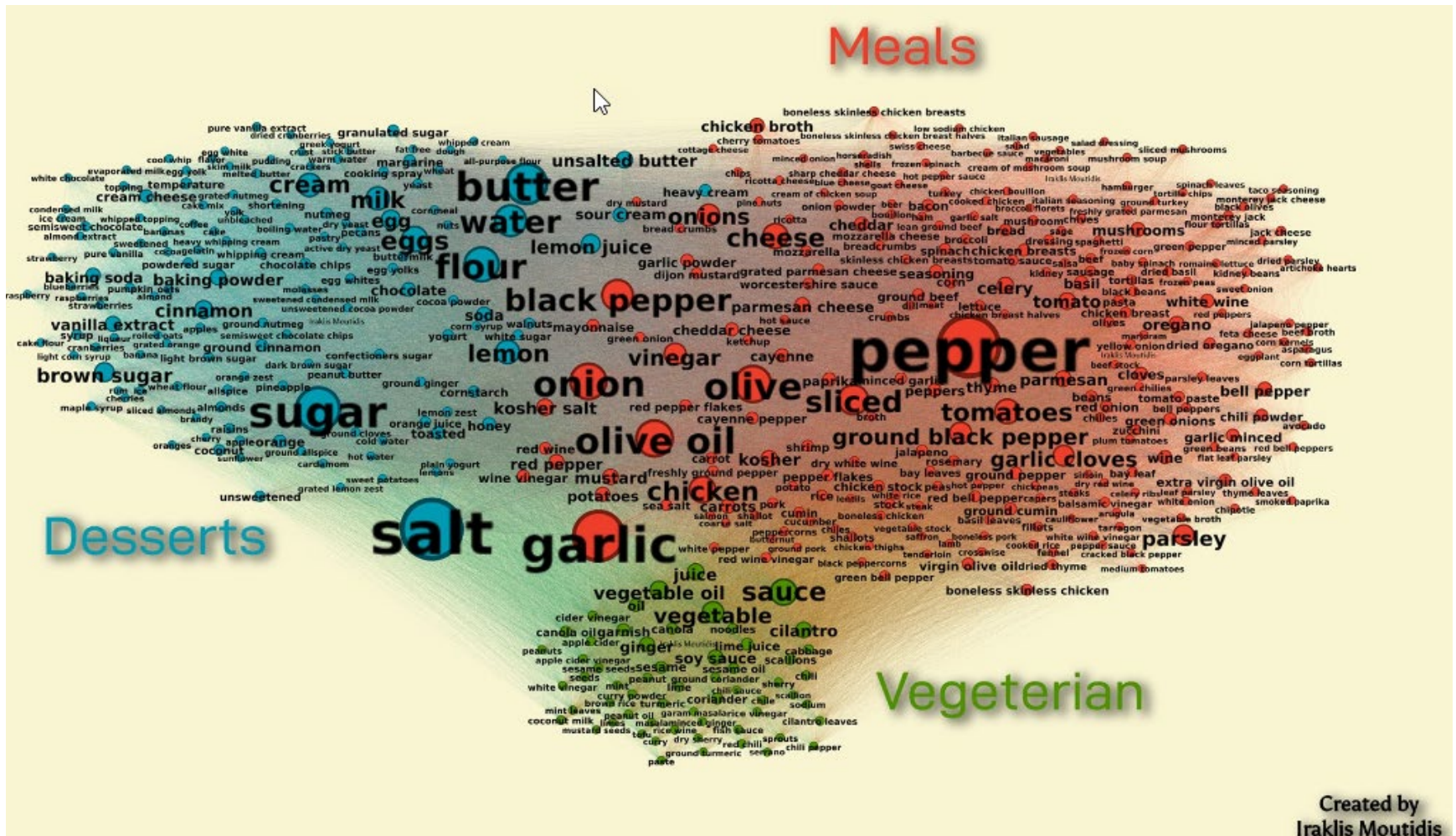
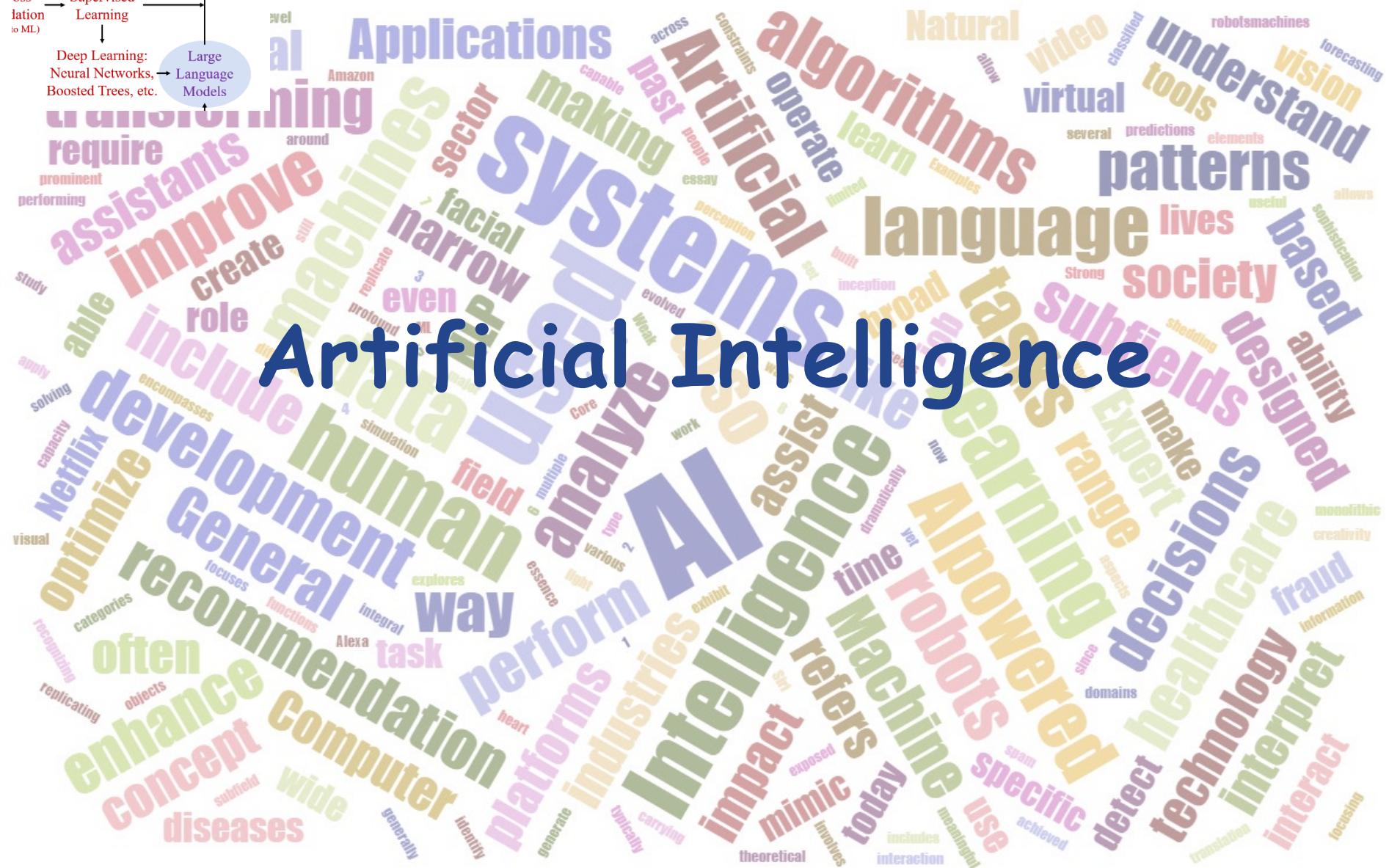
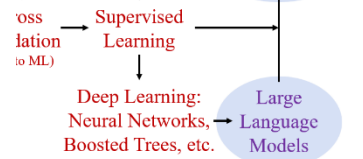


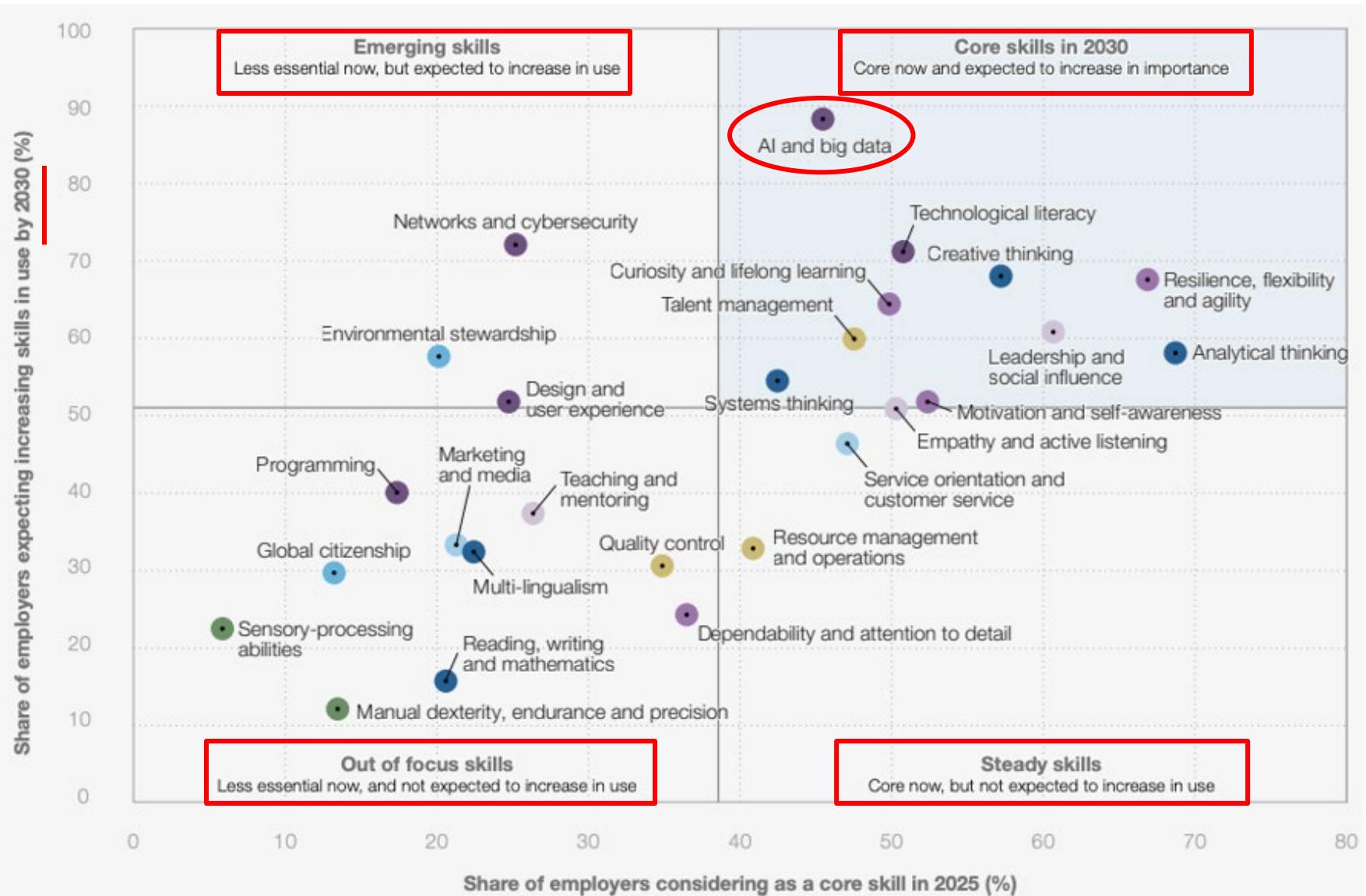
Illustration: Food Ingredient Network





Core skills in 2030

Share of employers considering skills to be a core skill in 2025 and share of employers expecting skills to increase in importance by 2030.



Source

World Economic Forum, Future of Jobs Survey 2024.
(> 14.1 M workers worldwide from 1,043 companies)

Note

The Future of Jobs Survey uses the World Economic Forum's Global Skills Taxonomy.
Bold lines represent the median values across all skills.

The Top 50 Gen AI Web Products, by Unique Monthly Visits

1.  ChatGPT	11.  removebg	21.  IIElevenLabs	31.  manus	41.  replit
2.  Gemini	12.  Doubao	22.  Hugging Face	32.  CIVITAI	42.  candy.ai
3.  deepseek	13.  NotebookLM	23.  Lovable	33.  KlingAI	43.  JuicyChat
4.  Grok	14.  SPICYCHAT.AI	24.  Crushon AI	34.  cutout.pro	44.  VEED
5.  character.ai	15.  SUNO	25.  GAMMA	35.  Adot	45.  Hailuo AI
6.  perplexity	16.  QuillBot	26.  CURSOR	36.  DeepAI	46.  Meta AI
7.  Claude	17.  Kimi	27.  Pixelcut	37.  Poe	47.  Remaker
8.  JanitorAI	18.  PolyBuzz	28.  Midjourney	38.  ZeroGPT	48.  Joi
9.  Quark	19.  SEARTE.AI	29.  TurboScribe	39.  Google Labs	49.  Monica
10.  Google AI Studio	20.  Qwen3	30.  Photoroom	40.  Leonardo.Ai	50.  ourdream.ai

Source: Similarweb, August 2025

Charts are for informational purposes only. Past performance is not indicative of future results. None of the above should be taken as investment advice; see a16z.com/disclosures.

Source: Andreessen Horowitz, August 2025

<https://a16z.com/100-gen-ai-apps-5/>

Artificial Intelligence (AI)

- Current consensus: **AI** will not **replace jobs**, people who know **AI** will !!
- **AI** is an **old** discipline – started in 1956 (Dartmouth Prof. John McCarthy)
- Defined as: *"the science and engineering of making intelligent machines that behave in ways that would be called intelligent if a human were so behaving"*
→ *Learning, Reasoning and Self-Correcting.*
- There are **many branches** of **AI**, such as:
 - ✓ **Logic-Based** – programs that follow rules, facts and human logic
 - ✓ **Expert Systems** – logic-based → decision rules elicited from experts
 - ✓ **Machine Learning (ML)** – *systems that learn mostly from historical data, without explicit programming*
 - ✓ **Generative AI (GAI)** – *“deep learning models that can generate high-quality text, images and other content based on data they were trained on”* (IBM) → **generate** content from user **prompts**.

Current Trends in AI

- **Large Language Models:** generate human-like text, images, etc. from words (tokens) you prompt and use **supervised learning** trained (with >1.5 trillion parameters) to predict the next word based on patterns it sees, over and over, sequentially.
- **Generative AI (GAI):** responds to prompts and generates responses as text, images, sound, video, etc.
- **Retrieval-Augmented Generation (RAG):** the prompt is first queried on **factual, specific,** and/or **proprietary information** (e.g., a company's product catalog or corporate policy), and the output is used to **augment** the original prompt sent to the GAI, thus improving quality and accuracy of the response, and reducing **hallucination**.
- **Agentic AI:** Programming (with code or visually) a process of running a prompt (ad-hoc or automatic) through a GAI tool, using the output to feed as input to another (or the same) GAI tool, then again multiple times to accomplish a goal (e.g., booking a vacation)
- **Multimodal Models:** Can process and combine data from different sources (text, image, video, audio, etc.) → Key role of **"transformers"**.

Superagency in the Workplace

McKinsey
& Company

- Survey of 3,613 employees and 238 C-level executives

<https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/superagency-in-the-workplace-empowering-people-to-unlock-ais-full-potential-at-work>

Superagency i.e., human performance augmented by AI in the Workplace

Empowering people to unlock AI's full potential

- 92% of companies plan to invest in AI over the next 3 years
- 1% believe investments have reached maturity
- Employees are more ready for AI than their leaders are or imagine

Hannah Mayer
Lareina Yee
Michael Chui
Roger Roberts

January 2025

Prompt Engineering

It is the process of designing, refining, and optimizing prompts when interacting with AI models like GPT (Generative Pre-trained Transformer).

Key aspects:

- **Clarity and Specificity** → ensure the model understands what's asked
- **Contextualization** → general context (e.g., who are you? what are your needs?)
- **Instructions for Tasks** → state what's expected (e.g., summary, detailed response)
- **2 Approaches:** (1) long, thorough prompt; or (2) short prompt & iterate

Prompt:

“I am a machine learning professor at a business school. How do I explain what prompt engineering is to an audience of young students?”

ChatGPT Response:

“To explain prompt engineering to an audience of young students, you can draw parallels between how employees optimize business processes and how prompt engineers optimize AI outputs. Here’s an example: Prompt engineering is the process of crafting precise, specific instructions for an AI model to ensure it provides accurate and relevant results, much like how you would design a financial query to get the correct data from a business database. In the same way that vague or poorly structured inputs can lead to incorrect human reports, unclear prompts can lead to less useful AI responses. By refining these prompts, we help the AI deliver exactly what we need, whether it’s a financial summary, analysis, or any other task.”

Prompt Engineering Example

Provided by an AI artist who creates images art in the style of known artists, describing what the image should contain along with key characteristics of the author's style.

Prompt (short version):

“In the muddy field, a man and a woman wearing traditional clothing stand near a wheelbarrow filled with hay. The man wears a brown jacket, while the woman is dressed in a mauve dress and a matching headscarf, seemingly in prayer. A picturesque backdrop features a landscape with a church, open fields, and a sky displaying hues of blue, orange, and purple, indicating either dawn or dusk. The pitchfork’s tines are thicker.”

Generated with [ideogram.ai](https://www.ideogram.ai), by [Arnold Millet](#)



Prompt Engineering Example

Same prompt, done by me with ChatGPT, but adding “in the style of Jean-Francois Millet”

Prompt:

*“In the muddy field, a man and a woman wearing traditional clothing stand near a wheelbarrow filled with hay. The man wears a brown jacket, while the woman is dressed in a mauve dress and a matching headscarf, seemingly in prayer. A picturesque backdrop features a landscape with a church, open fields, and a sky displaying hues of blue, orange, and purple, indicating either dawn or dusk. The pitchfork's tines are thicker. **Render the image in the style of Jean-Francois Millet.**”*

Generated with ChatGPT, by me, same prompt



Prompt Engineering Example

Original famous painting: “The Angelus” by Jean-François Millet

How would you improve upon the previous prompt to render a more accurate drawing?

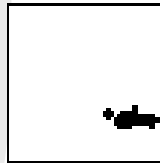


Final Words of Wisdom

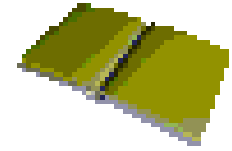
- **Learn AI** → Take AI courses and/or enroll in AI programs; AI won't take your job, but someone who knows AI will !! AI is here to stay. Find out how professionals in your **domain of interest** (e.g., accounting, finance) are using AI and learn their practices.
- **Learn to code** → Programming in Python & R is key.
- **The future** → Nobody can predict what you'll need to know in 10 years. What you'll learn today will probably not be in use in 10 years, but it will give you a knowledge base and help you learn future technologies.
- **Tech vs. Biz** → Figure out where your interests lie in the spectrum from **technical** (e.g., programming, data science, AI developer, etc.) to **business** (e.g., AI applications and uses, privacy, governance).
- **Personal use** → **Don't** ask AI to do the **work for you** !! → If AI can do your work, why would anyone hire you; AI can **hallucinate** → Use AI to **enhance** your work and productivity (e.g., writing, research, copilot).
- **Use responsibly and ethically** → there is great potential for unethical and harmful uses of AI → use AI for good and report unethical uses



Questions?



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Reference Information



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Kogod School of Business @ American University

Information Systems, Analytics & AI Specializations & Major

Business Analytics and Artificial Intelligence Major (BAAIm)

(Major: 38 cr. Business Core + 12 cr. BAAIm Core & Programming + 15 cr. BAAIm Electives & AI Applications)

The major is designed to provide students with advanced technical skills and business knowledge in a data- and AI-driven world. Business analytics is about extracting meaningful insights from data to inform decision-making and drive AI. AI is the science of making machines perform tasks that typically require human intelligence, such as learning, reasoning, and problem-solving. This major integrates core business foundations with in-demand skills in data analytics, machine learning, and the responsible use of AI. Students acquire hands-on experience transforming raw data into actionable insights, building and applying AI models, and designing intelligent business solutions. BAAI affects most areas of business (e.g., sports, healthcare, marketing, etc.). The BAAIm program also provides students with a strong foundation for business analytics, data management, analytics modeling, and software tools (e.g., R, Python).

Business Analytics and AI Specialization (BAAIs) (Specialization: 10 cr. Core + 15 cr; Minor: 18 cr.)

The specialization is for Kogod BSBA students, and the minor is for non-Kogod students. **BAAI** affects most areas of business (e.g., sports, healthcare, marketing, etc.). The curriculum is similar to the BAAIm, but with fewer course requirements.

Information Systems and Technology Specialization (IST) (Specialization: 10 cr. Core + 12 cr; Minor: 18 cr)

The specialization is for Kogod BSBA students and the minor is for non-Kogod students. **IST** education focuses on learning how IT can be best utilized and managed to innovate products and processes, transform firms, and create business value. **IST** encompasses technology, people, business processes, and business models. **IST** professionals serve as a bridge between business and IT. Students who understand the functional areas (e.g., marketing, finance, accounting, HR, etc.) but also have a solid understanding of how IT works and how it can support business processes and models are ideally suited for **IST** careers such as consultants, business analysts, cybersecurity specialists, and IT project managers.

Legend: (RK) Required for all Kogod students; (GE) General electives; Business Analytics & Artificial Intelligence Major: (BAAIm) (bold) required; (BAAI) Electives; **Business Analytics Specialization: (BAAIs) Required; (BAAIs) Electives; Information Systems & Technology Specialization: (IST) Required; (IST) Electives**

Core Courses (3 cr. each unless noted)	Pre-Requisites	Requirements
KSB 113 Navigating Business Technologies	None	RK
KSB 121 Data Analysis in Practice (2)	None	RK
KSB 224 Business Analytics	ITEC 200 or KSB 113	RK
KSB 301 Introduction to Operations Management (2)	ITEC 200 or KSB 113	RK
Required for Specializations & Major		
ITEC 300 Programming Tools for Analytics (R)	ITEC200 or KSB113; STAT204 or KSB121	<u>BAAIm</u> , BAAIs , IST
ITEC 321 Advanced Business Analytics with AI	ITEC 300; ITEC 320 or KSB 224	<u>BAAIm</u> , BAAIs , IST
ITEC 455 Business Process Analysis	ITEC 200 or KSB 113	<u>BAAIm</u> , BAAIs , IST
ITEC 470 Database and AI	ITEC 200 or KSB 113	<u>BAAIm</u> , BAAIs , IST
ITEC 400 Programming Tools for Analytics (Python)	ITEC 200 or KSB 113; ITEC 300	<u>BAAIm</u> , BAAIs , IST
Electives		
ITEC 334 Online Business Development (Python)	ITEC 200 or KSB 113	<u>BAAIm</u> , BAAIs , IST
ITEC 350 Digital Leadership	None	<u>BAAIm</u> , IST
ITEC 423 Social and Organizational Network Analytics	ITEC 200 or KSB 113; ITEC 300	<u>BAAIm</u> , BAAIs
ITEC 460 Business Intelligence	ITEC 200 or KSB 113; STAT204 or KSB121	<u>BAAIm</u> , IST, BAAIs
ITEC 466 Cybersecurity Risk Management	ITEC 200 or KSB 113	IST
ITEC 467 Cybersecurity Governance	ITEC 200 or KSB 113	IST
ITEC 468 Cybersecurity Analytics	ITEC 200 or KSB 113; STAT204 or KSB121	<u>BAAIm</u> , IST, BAAIs
ITEC 477 Microsoft Certifications	ITEC 200 or KSB 113	GE
DATA-312 Data Science Applications		<u>BAAIm</u>

BS/MS Dual Degree in Analytics and AI → You can **double count** up to 9 credits 6XX for 4XX, or 5XX:

ITEC 623 for ITEC 423; ITEC 630 for ITEC 455; ITEC 660 for ITEC 460; ITEC 670 for ITEC 470; ITEC 666 for ITEC 466;
ITEC 667 for ITEC 467; ITEC 668 for ITEC 468; ITEC 5XX; Others as approved by MS Program Director

American University, Kogod MS Analytics

KSB 620 & 621 Capstone Experience Practicum (2 x 1.5 cr)

Electives & Functional Specialization (12 cr)

- ITEC 643 Project Management
- ITEC 630 Business Process Analysis
- ITEC 623 Organizational & Social Network Analytics
- ITEC 724 Advanced Text Analytics w/AI and Machine Learning
- ITEC 668 Cyber Security Analytics

Plus, specializations in:

- Cybersecurity
- Accounting Forensics
- Quantitative Financial Analysis I and II
- Marketing
- Accounting
- Data Science, etc.

Analytics Core (9 cr)

ITEC 610
Managerial Statistics

ITEC 620 Business Insights
Through Analytics

ITEC 621
Predictive Analytics
w/Machine Learning

IT Core (9 cr)

ITEC 616
Managing the Digital
Organization

ITEC 670
Database & Big Data

ITEC 660
Business Intelligence

Tools:
KSB 999 R;
ITEC 600 Python

BS/MS Option → Double-count 9 credits

Further details: <https://jibe4fun.github.io/analytics/msanalytics.pdf>



A Few Useful General AI Tools

- **GhatGPT** (<https://chatgpt.com/>): conversational **G**enerative **P**re-Trained **T**ransformer AI model. GPT-3.5 is free, ChatGPT Plus is \$20/month.
- **Perplexity** (<https://www.perplexity.ai/>): Conversational AI, similar to ChatGPT, but very accurate and yields useful citations
- **GROK** (<https://grok.com/>): Conversational AI chatbot developed by xAI, similar to ChatGPT
- **Copilot** (<https://www.microsoft.com/en-us/microsoft-copilot/personal-ai-assistant>): Microsoft AI tool with free and paid versions.
- **Gemini Notebook** (<https://notebooklm.google.com/>): By Google, good for generative AI on local documents like Word, Powerpoint and PDF. Was called NotebookLM before.
- **Gemini** (<https://gemini.google.com/app>): Gemini's chat tool with free and paid versions.
- **Dall-E** (<https://openai.com/index/dall-e-3/>): generates graphs in ChatGPT
- **MidJourney** (<https://www.midjourney.com/home>): Generates images from text.
- **Pearl.ai, AI Code Editor, Claude, Cursor**: generative AI tools with advanced features for software programming.
- **DeepL Translate** (<https://www.deepl.com/en/translator>): for accurate translation between multiple languages
- **Grammarly** (<https://www.grammarly.com/>): Excellent AI writing tool

Artificial Intelligence (AI) & Machine Learning Applications by Industry

- **Healthcare:** diagnostics, personalized treatment, robotic surgery, AI-driven diagnostic tools, drug discovery, predictive analytics for patient management.
- **Finance & Accounting:** risk management, fraud detection, accounting forensics, algorithmic trading, personalized banking services, and AI-driven compliance.
- **Retail:** personalized marketing, inventory management, sales forecasting, e-commerce, chatbots, recommendation engines.
- **Manufacturing:** production line optimization, predictive maintenance, quality control, automation.
- **Transportation:** self-driving cars, route optimization, logistics, traffic management, autonomous systems.
- **Energy:** energy use optimization, predicting equipment failure, managing grids, efficient energy consumption.
- **Advertising & Media:** content creation, targeted advertising, media analytics

Popular Uses of Generative AI

- **Research** → Search papers, summarize articles, find citations, etc.
- **Writing** → Write first drafts, edit writing, change styles (e.g., academic)
- **Local Knowledge Base** → Chat and generate from your own materials
- **Local Coach/Tutor** → Support learning and content creation
- **Advanced Searches** → Generate content, not just website links
- **Create Presentations** → Generate slides and graphics
- **Generate Creative Content** → Images, Music, Video, etc.
- **Software Programming** → Learn how to code, generate code, etc.
- **Language Translation** → Text or audio from/to any language
- **Power Computer User** → Tools (ex., Copilot for Excel, PowerPoint)
- **Voice Applications** → Voice assistants, transcriptions
- **Face Recognition** → Tagging photos, device security, etc.